AI-Powered Tour Guide Booking System with Voice and RAG

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SLIIT
FACULTY OF COMPUTING

AI-Powered Tour Guide Booking System with Voice and RAG

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PP2 Comments and Actions Taken

Topic	AI-Powered Tour Guide Booking System with Voice and RAG			
Project ID	25-26J-150			
Specialization	ISE			
Student ID	IT21191060	no	no	no
PanelName	ISE-P1			
Detailed Comments	The application requires context relavance and fine tuning of the algorithms used. the chat does not provide the itinerary details and does not identify the content related to the trip requirements.			

- Fine tuning of the algorithms.
- Itinerary details related to the trip requirements

Research Problem

- Tourists face difficulty in finding trusted, verified guides
- Existing platforms lack multi-level verification, leading to fraud and scams.
- Poor personalization and static recommendations don't consider culture, preferences, or real-time data.

Specific Objectives

- Develop a secure AI-powered web application for tour guide bookings that integrates multi-level document verification.
- Implement an intelligent matching and recommendation engine using Retrieval-Augmented Generation (RAG) to provide personalized, multilingual, and context-aware guidance for tourist

Sub Objectives

- Design user interfaces and workflows for tourist voice queries, guide registration uploads booking inputs.
- Integrate fraud detection and multilingual chat agent features to enhance trust and improve real-time tourist-guide interactions.

User Requirements addressed by the solution

- Admin: Tour guides accept or reject the Tour guides
- Tour guides **:** NIC, Police Report and SL Tourism card to upload and accept the trip
- Tourist : Assign the trip in the system by agent.

Functional Requirement

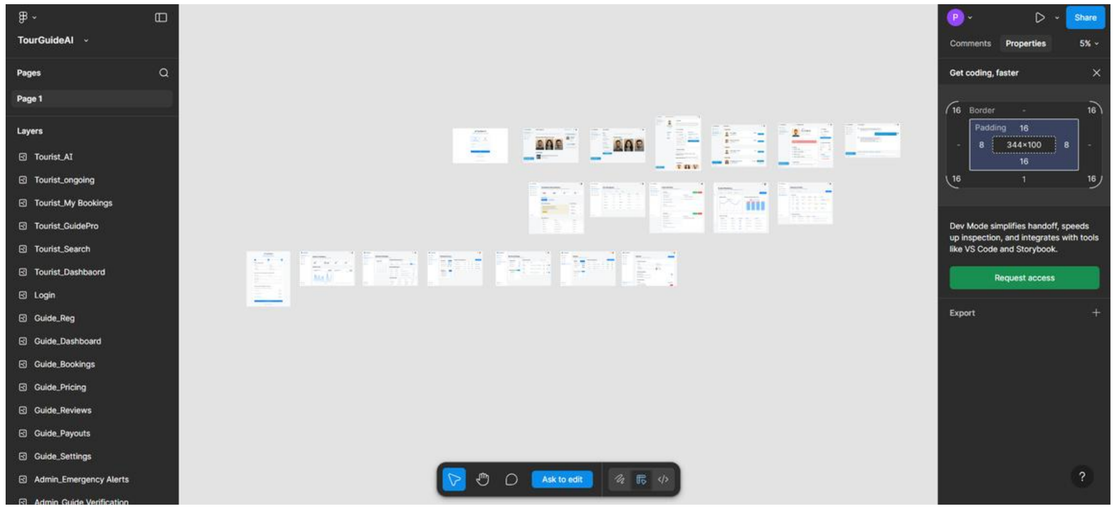
- User Registration and Authentication.
- Multi Document Verification – Validation of NIC, passport, licenses, and police check to categorize guides as verified or non-verified.
- Tourist Guide Matching – AI-powered RAG-based engine to recommend guides based on tourist preferences, location, and real-time factors.
- Multilingual AI Chat agent – Voice and text-enabled assistant supporting Sinhala, Tamil, and English for bookings, recommendations, and inquiries.

Non-Functional Requirement

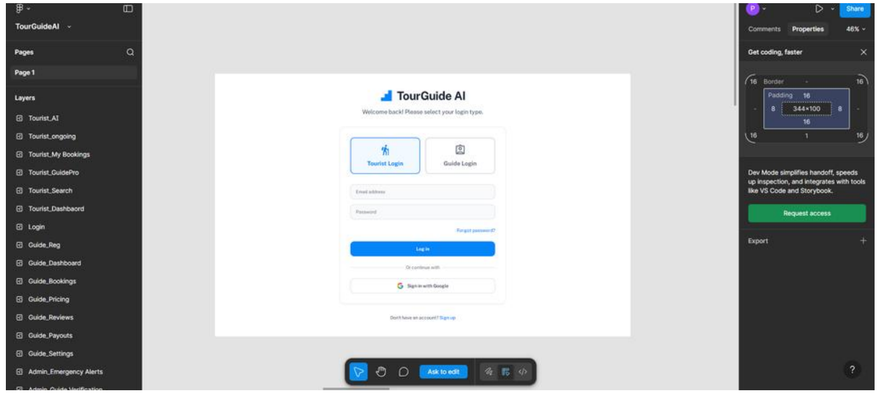
- Security
- Scalability
- Performance
- Reliability
- Cross-Platform Accessibility
- Usability

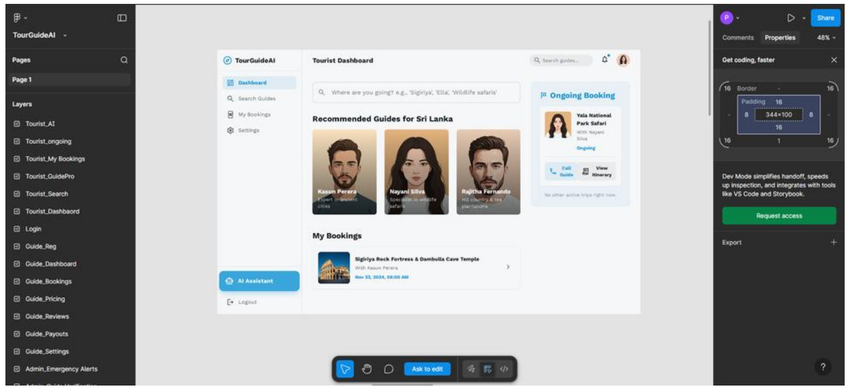
•Design Excellence/Contribution

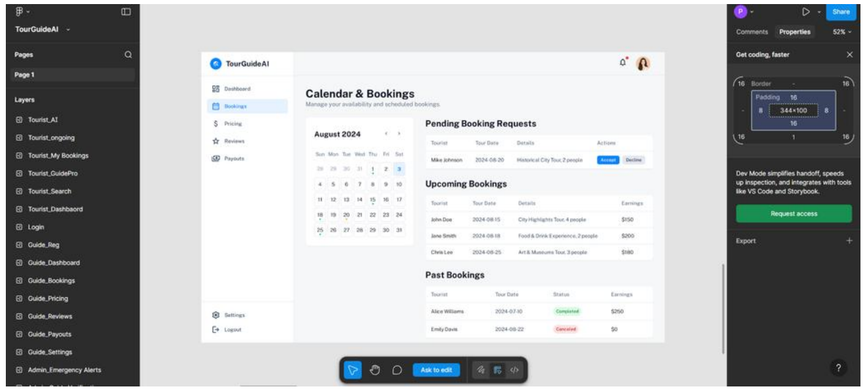
- Used FIGMA for prototype

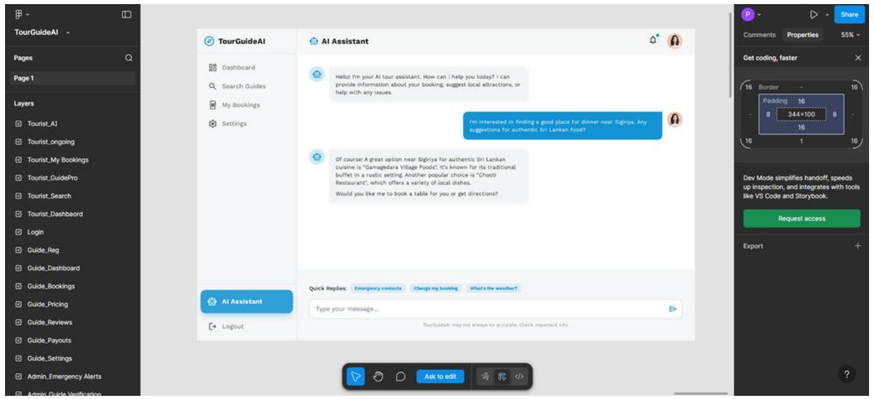


•Design Excellence/Contribution

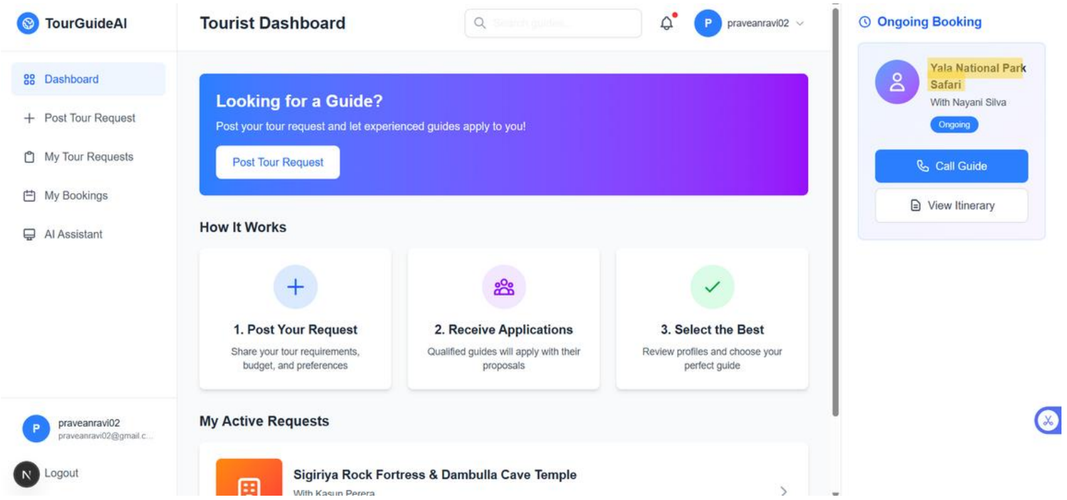


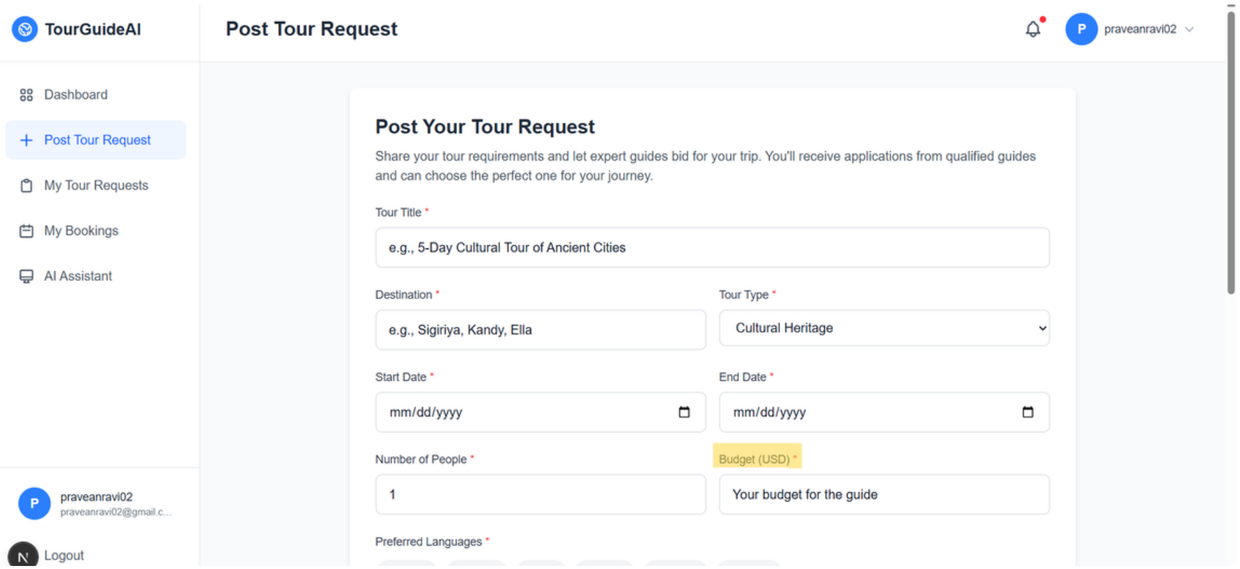






User Feedback on Prototype

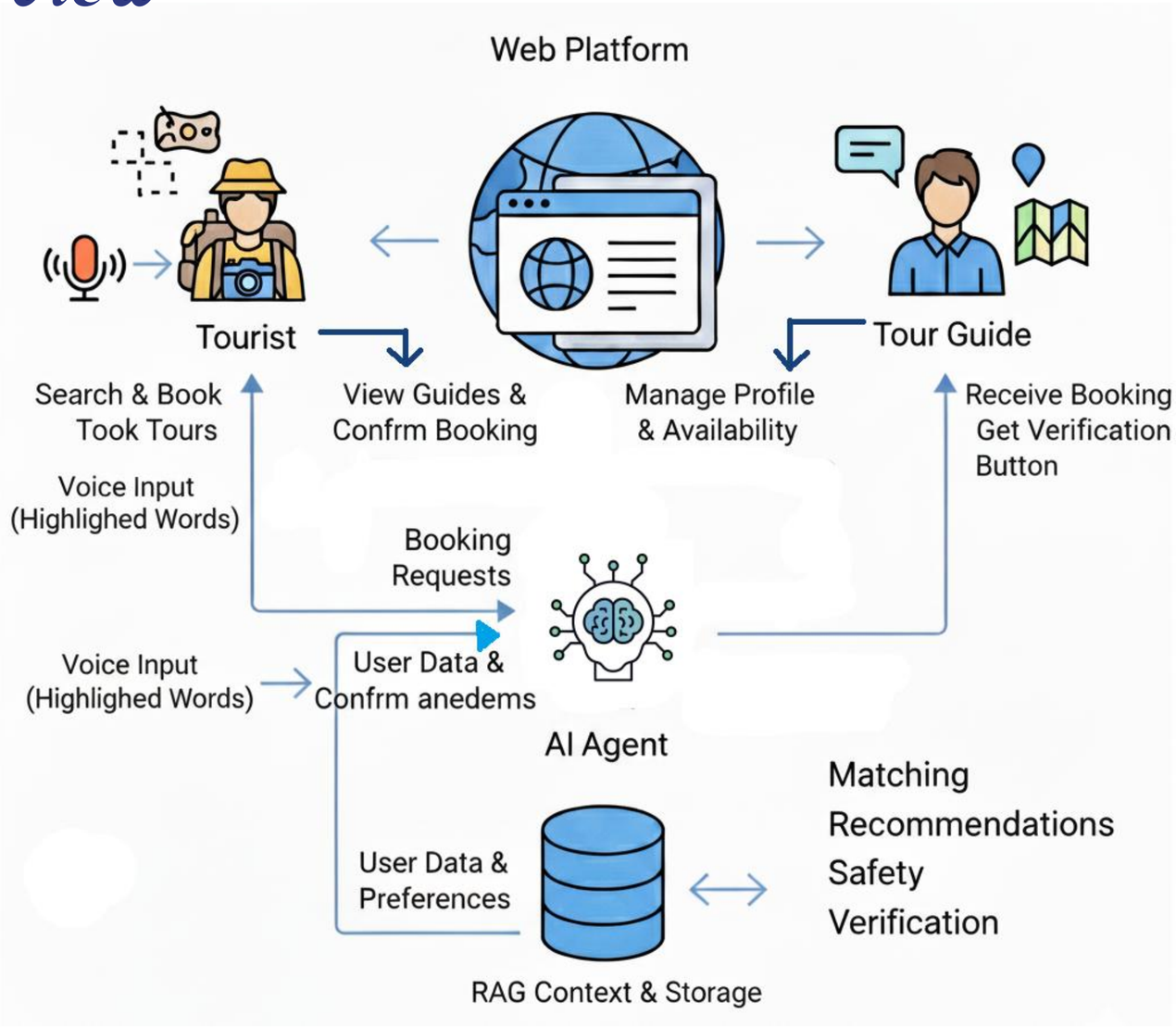




Technologies

- Frontend: React.js for interactive UI.
- Backend: Node.js/Express with secure API integrations and Python.
- Database: Vector DB for RAG retrieval + firebase DB.
- AI/ML: NLP, keyword extraction, RAG for intelligent responses.
- Multilingual Support: Sinhala, Tamil, English chat agent voice interface.

System Overview



Commercialization

- Commission-based revenue from bookings.
- Subscription plans for verified guides (premium listings).
- Partnerships with hotels, airlines, and travel agencies.

Target Audience and Market Size

- Primary: local tourists seeking and verified guides.
- Secondary: Professional tour guides seeking visibility and trust.
- Tertiary: Tourism boards, hotels, and travel agencies.

Investment / Cost & Recovery

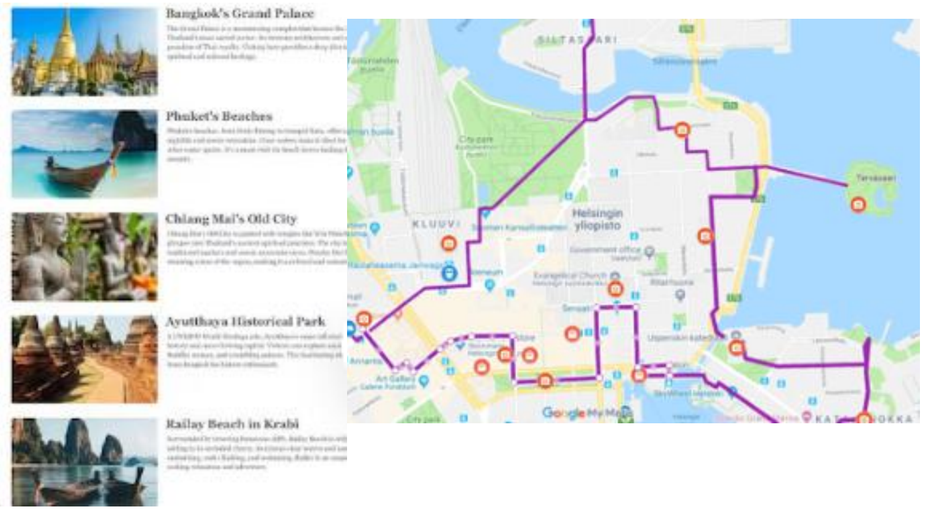
- Commission fees (10–15%) per booking.
- Subscription fees from guides.
- Sponsored placements from travel partners.

What is Unique in Your Solution

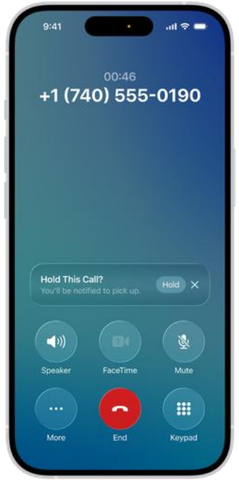
- For Tourists: Safety, trust, multilingual AI support, personalized journeys.
- For Guides: Verified exposure, fair marketplace, protection from fraud.
- For Sri Lanka: Enhanced tourism reputation, repeat visitors, digital innovation leadership.

Limitations of the System and Future Improvements

- Uable to add images when showing itinerary details .

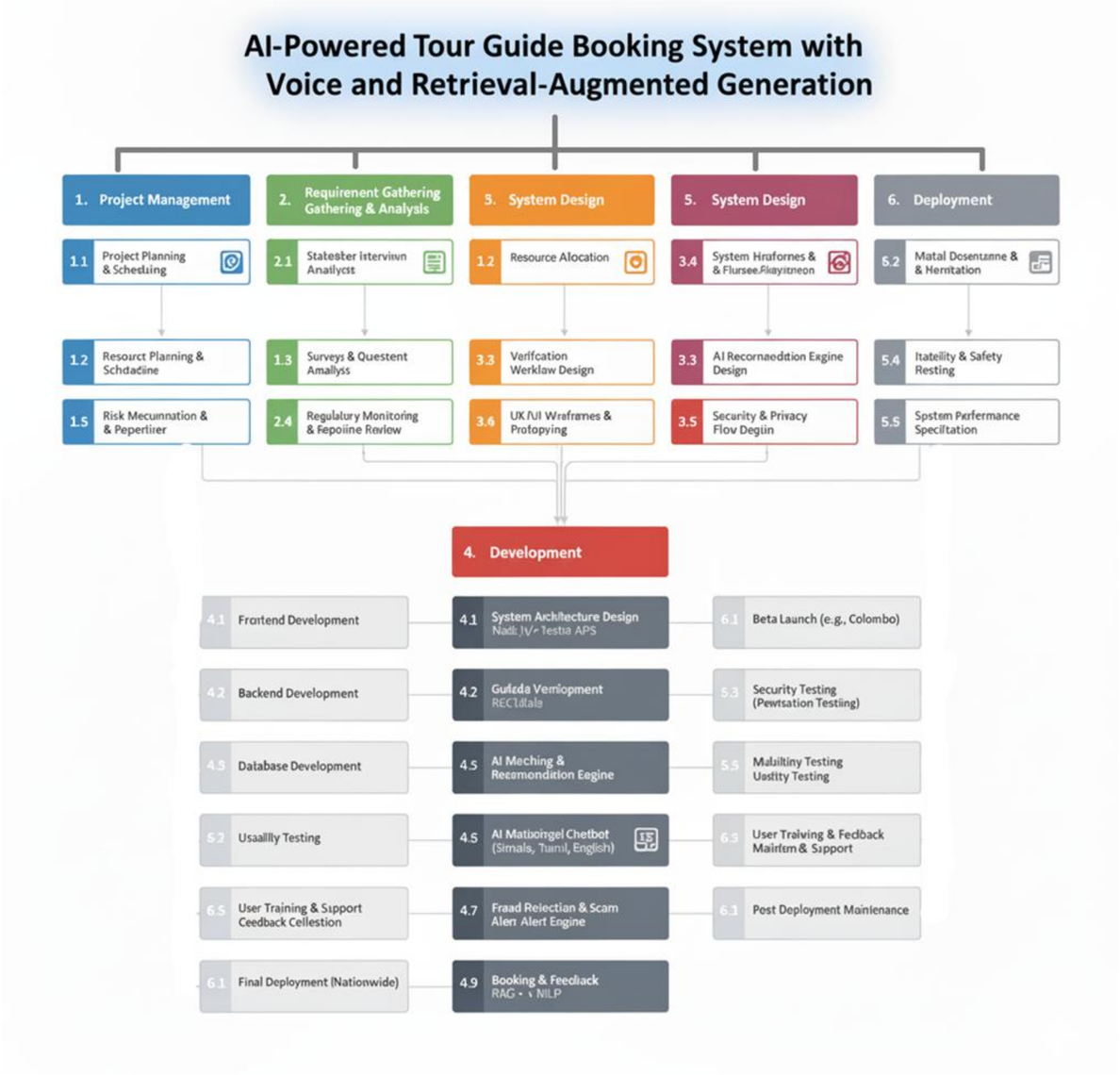


- Uable to call tourguied in trip via website .



- Website next into Mobile app.

WBS


AI-Powered Tour Guide Booking System with Voice and Retrieval-Augmented Generation
1. Project Management
2. Requirement Gathering Gathering & Analysis
3. System Design
5. System Design
6. Deployment
1.1 Project Planning & Scheduling
2.1 Stakeholder Interviews Analysis
1.2 Resource Allocation
3.4 System Hrafornes & & Flursee.Fkayitneon
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1.2 Resourct Planning & Schdaciine
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3.3 Verifcation Werklaw Design
3.3 AI Recornaedtion Exgine Design
5.4 Itateility & Safety Resting
1.5 Risk Mecumnation & & Pepertiier
2.4 Regulatory Monitoring & Fepoiine Review
3.6 UX/UI Wraframes & Protopying
3.5 Security & Privacy Flov Degiin
5.5 Spstem Parfermance Speciltation
4. Development
4.1 Frontend Development
4.1 System Architecture Design Nadi.JV- Testia APS
6.1 Beta Launch (e.g., Colombo)
4.2 Backend Development
4.2 Guideda Vemiopment RECldlala
5.3 Security Testing (Pewtsation Testiing)
4.5 Database Development
4.5 AI Meching & Recomondition Eegine
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5.2 Usaaility Testing
4.5 AI Matboingel Chetbot (Simals, Tuinil, English)
6.5 User Traiving & Fedback Mairfem & Sipport
6.5 User Training & Sipport Ceedback Cellestion
4.7 Fraad Reiectian & Scam Alert Alert Engine
6.1 Post Deployment Mainlenance
6.1 Final Deployment (Nationwide)
4.9 Booking & Feedliack RAG + NILP

References

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- G. Boza-Quispe, J. Montalvan-Figueroa, J. Rosales-Huamaní and F. Puente-Mansilla, "A friendly speech user interface based on Google cloud platform to access a tourism semantic website," 2017 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON), Pucon, Chile, 2017, pp. 1-6.

Thank You